

Unified Flow: clocks, entropy and spectral probes

Complete calculation notes and source-linked proof records

12 September 2026

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Chapter 1

A specified black-hole calculation

The **Unified Flow** workbench tests proposed relations among physical clocks, thermal states, detector response, entropy, spectral transforms and geometric flow. Its first worked system is an eternal Schwarzschild black hole in a Hartle-Hawking thermal state. Each probe, unit, clock and operator is retained: the goal is to compute the maps between them, not to declare all flows equal.

The full source calculation follows. It distinguishes a four-dimensional scalar scattering problem, a conformal two-dimensional detector, a Dirac shell probe and an interacting sigma-model probe. The detector's energy scale, thermal clock, entropy variation and spectral-zeta transform are computed for their specified models. These are related calculations within the wider research workbench, not a claim of solving quantum gravity or RH.

The subsequent chapters retain the computational checks and contributor/source history. Source literature is cited rather than bundled into this edition.

Chapter 2

Unified Flow with a black hole in the equations

[Complete source note](#). Module: `workbenches/unified-flow/HARTLE_HAWKING.md`.

Revision: [512fb1a6aa0922318aff49081c01674cf30ec906](#). Complete selected note.

11 September 2026. Working derivation and research interfaces; not an independent mathematical review.

1. Physical systems and source history

Choose an eternal, uncharged, nonrotating four-dimensional Schwarzschild black hole of mass $M > 0$, in Einstein gravity with Newton constant G . A static laboratory is held at areal radius $R > r_h$. Quantum fields are treated as test fields in Hartle-Hawking equilibrium. This has incoming as well as outgoing thermal excitations, not the state of an isolated evaporating black hole. Backreaction of the entire equilibrium bath is not being solved. Comparisons between nearby stationary masses below are separate variations. [1-3]

Keep three probes distinct:

- A free, real, minimally coupled massless scalar in four dimensions, measured by a monopole Unruh-DeWitt detector. Its actual radial scattering problem is specified in Section 5.
- A massless conformal scalar on the two-dimensional radial metric, measured by a derivative-coupled two-level detector. This explicitly omits the angular potential and the dilaton factors of literal spherical reduction. Its peak is exactly calculable. [4]
- A massless complex Dirac field confined to a round test shell of radius R , propagation speed c , zero chemical potential and the round-sphere spin structure. Populate its positive particle and antiparticle excitations at the local equilibrium temperature. This is a thermal surface probe, not an asserted microscopic model of the horizon.

Section 9 adds a separately identified interacting sigma-model probe rather than inventing an interaction beta function for the free Dirac field.

Historical sources: `206.tex`, lines 1585-1709, contains the detector-to-scale calculation; A figment of `acceleration.pdf`, extracted text lines 212-276, proposes the Hartle-Hawking/modular/spectral-entropy application; OLDER notes that have

stuff on Ricci flow.txt, lines 67-107, contains Schwarzschild thermodynamic comparisons. The old entropy-quantization postulate is not assumed. Those originals are not republished here, and their theorem headings are not verification certificates. CONTRIBUTIONS.md records the user-attested Gemini 2.5 Pro origin of modular action.

2. Geometry, thermal state, and exact clock conversions

Retain all dimensional constants. Put

$$r_h = \frac{2GM}{c^2}, \quad f(r) = 1 - \frac{r_h}{r}, \quad N = \sqrt{f(R)}, \quad \ell_P^2 = \frac{G\hbar}{c^3}.$$

The metric and laboratory clock are

$$ds_{\text{geom}}^2 = -f(r)c^2dt^2 + f(r)^{-1}dr^2 + r^2d\Omega_2^2, \quad d\tau_R = Ndt.$$

Contraction of a momentum with the Killing field and the static unit tangent gives the energy map

$$E_\infty = NE_R.$$

Surface gravity in acceleration units, with Killing time normalized at infinity, is

$$\kappa = \frac{c^2 f'(r_h)}{2} = \frac{c^2}{2r_h} = \frac{c^4}{4GM}.$$

Euclidean regularity fixes the thermal circle:

$$\hbar\beta_H = \frac{2\pi c}{\kappa} = \frac{4\pi r_h}{c}, \quad k_B T_H = \frac{\hbar c}{4\pi r_h}.$$

The proper thermal period is redshifted by N , so

$$\boxed{\beta_R = N\beta_H, \quad T_R = T_H/N, \quad \beta_R E_R = \beta_H E_\infty.}$$

These are the Tolman/KMS relations for the chosen state. [1-3]

The accelerometer reading follows from $u^\wedge_t = 1/N$ and $\Gamma^\wedge_{r_{tt}} = c^2 f'/2$:

$$a_{\text{stat}} = \frac{c^2 f'(R)}{2N} = \frac{GM}{R^2 N}.$$

Define the acceleration-valued thermal scale and retain the conversion factor:

$$a_{\text{KMS}} := \frac{2\pi c k_B T_R}{\hbar} = \frac{\kappa}{N}, \quad \boxed{a_{\text{stat}} = (r_h/R)^2 a_{\text{KMS}}.}$$

The ratio tends to one near the horizon and is 1/4 at $R=2r_h$. Thus the local Rindler limit is visible without replacing the exact Schwarzschild relation.

The proper distance on a static slice is

$$\rho(R) = \sqrt{R(R-r_h)} + r_h \log \frac{\sqrt{R} + \sqrt{R-r_h}}{\sqrt{r_h}}, \quad \frac{d\rho}{dR} = \frac{1}{N}.$$

For $\delta = R - r_h$ small,

$$\rho = 2\sqrt{r_h\delta} \left[1 + \frac{\delta}{6r_h} + O(\delta^2/r_h^2) \right], \quad k_B T_R = \frac{\hbar c}{2\pi\rho} [1 + O(\rho^2/r_h^2)].$$

The leading local temperature at a fixed proper distance is independent of M .

3. The modular action in this physical state

Let α_t be Killing-time evolution and fix

$$\sigma_{s_{\text{TT}}} = \alpha_{-\hbar\beta_H s_{\text{TT}}}.$$

Retain both the reversed coordinate $u = -s_{\text{TT}}$ and its action-valued counterpart $J = \hbar(u - u_0)$. Then

$$\boxed{\frac{du}{dt} = \frac{1}{\hbar\beta_H}, \quad \frac{du}{d\tau_R} = \frac{1}{\hbar\beta_R}, \quad \frac{dJ}{d\tau_R} = k_B T_R, \quad \frac{dJ}{dt} = k_B T_H.}$$

This measures thermal energy scale per unit proper time. Comparison using the same elapsed Killing time removes the laboratory radius. It is not heat generation: the equilibrium state is invariant under its own modular dynamics.

For finitely many exterior modes, take $H_{\infty} = \sum \epsilon_j n_j$ and $\rho = \exp(-\beta_H H_{\infty})/Z$. Then $K = -\log \rho = \beta_H H_{\infty} + \log Z$; substitution into $\rho^{(\text{is})}$ A $\rho^{(-\text{is})}$ proves the clock relation, with the additive $\log Z$ retained in entropy. The continuum exterior uses the algebraic KMS formulation, not a fictitious trace-class density matrix for a type-III local algebra.

4. A closed-form black-hole detector channel

For the conformal radial scalar, use

$$r_* = r + r_h \log(r/r_h - 1), \quad w_- = t - r_*/c, \quad w_+ = t + r_*/c, \\ U = -L e^{-\kappa w_-/c}, \quad V = L e^{\kappa w_+/c}.$$

With retained field coefficient A_ϕ , the HHI two-point function is

$$W(x, x') = -A_\phi \log \frac{(U - U' - i0)(V - V' - i0)}{L_0^2}.$$

The additive infrared constant disappears after derivative coupling. At fixed R let $p_R = \kappa/(cN) = 2\pi/(\hbar\beta_R)$. Direct differentiation gives

$$\partial_\tau \partial_{\tau'} W = -\frac{A_\phi p_R^2}{2} \text{csch}^2 \frac{p_R(\tau - \tau' - i0)}{2}.$$

For $H_{\text{int}} = g_d \chi_m d\phi/d\tau$, stationary leading-order perturbation theory yields

$$\mathcal{R}_+(\omega) = C_d \frac{\omega}{e^{\hbar\beta_R \omega} - 1}, \quad C_d = \frac{4\pi A_\phi g_d^2 |m_{eg}|^2}{\hbar^2}.$$

The Fourier integral is

$$\int_{\mathbb{R}} e^{-i\omega s} \text{csch}^2 \frac{p_R(s - i0)}{2} ds = -\frac{8\pi\omega}{p_R^2 (e^{2\pi\omega/p_R} - 1)}, \quad \omega > 0.$$

Closing clockwise in the lower half-plane sums the double poles at $-2\pi i n/p_R + i0$, $n \geq 1$. The residues form the geometric series giving the displayed sign and coefficient. [4]

Absorbed power in a scanned family of initially ground-state detectors with fixed C_d is

$$P_R(\omega) = C_d \frac{\hbar \omega^2}{e^{\hbar \beta_R \omega} - 1}.$$

The invertible coordinate $x = \hbar \beta_R \omega$ gives the positive maximum condition

$$(2 - x)e^x = 2, \quad k_* = 2 + W_0(-2e^{-2}) = 1.593624260040040 \dots$$

Uniqueness follows because $2(1 - e^{-x}) - x$ has positive initial derivative, strictly negative second derivative, and becomes negative; the power tends to zero at both endpoints. The other real Lambert branch gives the endpoint $x=0$.

Consequently

$$\mu_R = \frac{k_*}{\beta_R} = \frac{k_* \hbar c}{4\pi r_h N}, \quad \mu_\infty = N \mu_R = \frac{k_* \hbar c}{4\pi r_h}.$$

This is not equilibrium net detector power: excitation and de-excitation balance once the detector equilibrates. It is not a universal peak for every black-hole detector or for outgoing Hawking luminosity.

5. The four-dimensional field and its remaining numerical calculation

The original four-dimensional scalar has unit-Klein-Gordon-normalized in/up modes satisfying

$$\frac{d^2 u_{\nu\ell}}{dr_*^2} + \left[\frac{\nu^2}{c^2} - f(r) \left(\frac{\ell(\ell+1)}{r^2} + \frac{r_h}{r^3} \right) \right] u_{\nu\ell} = 0.$$

The in solution is incoming from past null infinity; the up solution is incoming from the past horizon. Their incoming KG flux fixes normalization. Let $J_R(\nu)$ be the sum over their squared normalized amplitudes, including ℓ, m , at the laboratory. This is a specified scattering boundary-value problem. [3]

$$W_R(s) = \int_0^\infty J_R(\nu) \left[(1 + n_H(\nu)) e^{-i\nu s/N} + n_H(\nu) e^{i\nu s/N} \right] d\nu,$$

$$n_H(\nu) = (e^{\hbar \beta_H \nu} - 1)^{-1}.$$

For monopole coupling,

$$\mathcal{R}_{+,4}(\omega) = \frac{2\pi g_d^2 |m_{eg}|^2 N}{\hbar^2} \frac{J_R(N\omega)}{e^{\hbar \beta_R \omega} - 1}.$$

A smooth positive power maximum obeys

$$\frac{1}{\omega} + N \frac{J'_R(N\omega)}{J_R(N\omega)} = \frac{\hbar \beta_R}{1 - e^{-\hbar \beta_R \omega}}.$$

A linear density recovers the k_* equation. Curvature scattering can alter it. This pass has not computed J_R numerically. Nevertheless the exact detailed-balance ratio fixes the same thermal clock:

$$\mathcal{R}_{+,4}(E)/\mathcal{R}_{-,4}(E) = e^{-\beta_R E}.$$

Thus the physically meaningful thermal measurement survives without assuming the flat-space peak.

6. Carry the selected scale into horizon entropy and area

Use Einstein black-hole entropy, not single-mode entropy:

$$S_{\text{BH}} = \frac{k_B A_H}{4\ell_P^2}, \quad A_H = 4\pi r_h^2, \quad \sigma := S_{\text{BH}}/k_B = \frac{4\pi G M^2}{\hbar c}.$$

With $E_P = \sqrt{\hbar c^5/G}$,

$$\mu_\infty = \frac{k_* E_P}{4\sqrt{\pi\sigma}}, \quad \mu_R = \frac{k_* E_P}{4N\sqrt{\pi\sigma}}, \quad \frac{dJ}{dt} = \frac{E_P}{4\sqrt{\pi\sigma}}.$$

Larger entropy means lower thermal energy and slower modular-action rate at infinity. No arbitrary cutoff occurs in this relation; k_* retains the specified detector observable.

Direct differentiation gives $T_H dS_{\text{BH}} = c^2 dM$. If a small neutral energy ϵ is absorbed and retained by the hole,

$$\delta S_{\text{BH}}/k_B = \beta_H \epsilon.$$

For $\epsilon = k_*/\beta_H$,

$$\delta S_{\text{BH}}/k_B = k_*, \quad \delta A_H = 4k_* \ell_P^2.$$

This is a first-law response, not an area quantum. Actual absorption is required; the existence of a detector response is not a flux into the hole.

The finite stationary-endpoint difference is also explicit:

$$\frac{S(M + \epsilon/c^2) - S(M)}{k_B} = \beta_H \epsilon + \frac{4\pi G \epsilon^2}{\hbar c^5}.$$

Using the initial peak energy gives

$$\Delta S_{\text{BH}}/k_B = k_* + k_*^2/(4\sigma).$$

For a finite faithful thermal reference state, relative entropy gives $D(\rho||\rho_0) = \beta_H \Delta S_{\text{field}}/k_B$. Its first variation vanishes at $\rho = \rho_0$. This makes the field entanglement first law and the horizon first law share the same modular-energy variation; it does not identify their absolute entropies.

7. A specified gravitational action

The equilibrium Schwarzschild Euclidean action, including its asymptotic boundary subtraction, obeys

$$I_E/\hbar = \beta_H M c^2 - \sigma = \sigma,$$

using the Smarr identity $M c^2 = 2 T_H S_{\text{BH}}$. During one thermal Killing-time interval $\Delta t = \hbar \beta_H$, the modular-action clock accumulates $\Delta J = \hbar$. Hence

$$I_E = \sigma \Delta J_{\text{thermal}}.$$

Entropy, rather than a coefficient chosen to be one, is the proportionality. No Wheeler-DeWitt CA identity is asserted.

For a Euclidean opening angle $2\pi n$, the integrated conical curvature is $4\pi(1-n)A_H$. Multiplication by the Einstein coefficient $-c^2/(16\pi G)$ gives

$$I_E(n) = nI_E(1) + (n-1)\frac{\hbar A_H}{4\ell_P^2} + O((n-1)^2).$$

Therefore

$$(n\partial_n - 1)\frac{I_E(n)}{\hbar}\bigg|_{n=1} = \frac{A_H}{4\ell_P^2}.$$

This is the semiclassical conical/replica entropy map. Terms linear in the circle length cancel. Renormalized matter adds the corresponding field contribution to generalized entropy, rather than allowing an arbitrary field entropy to stand in for the gravitational term. [5]

8. Fully specified fermion: spectrum, purification, entropy, zeta

The round-shell spectral triple is

$$\mathcal{A} = C^\infty(S^2), \quad \mathcal{H} = L^2(S^2, S; R^2 g_{\text{unit}}), \quad D_R = R^{-1} D_1.$$

Its self-adjoint intrinsic Dirac eigenvalues are plus/minus n/R , $n \geq 1$, multiplicity $2n$ per sign. [6] In the angular-momentum representation $D_R = (\sigma \cdot L_{\text{ang}}/\hbar + 1)/R$, where $L_{\text{ang}} = -i\hbar r \times \text{grad}$. Coupling orbital angular momentum with spin $1/2$ gives those eigenvalue branches.

Fill the negative sea as a reference and use positive particle/antiparticle excitations of one complex massless Dirac field. The one-particle excitation operator $H_{1,R} = \hbar c |D_R|$ has energy $E_{n,R} = \hbar c n/R$ with multiplicity $4n$. This fixes the thermal excitation vacuum energy as reference zero; it does not erase a Casimir contribution to gravitational stress.

$$x := \frac{\beta_R \hbar c}{R} = \frac{4\pi r_h N}{R}, \quad Z_R = \prod_{n \geq 1} (1 + e^{-xn})^{4n}, \quad p_n = (1 + e^{-xn})^{-1}.$$

The logarithm of Z converges exponentially for every $R > r_h$. A mode purification and its actual fermionic canonical transformation are

$$|\text{TFD}_n\rangle = \frac{|00\rangle + e^{-xn/2}|11\rangle}{\sqrt{1 + e^{-xn}}},$$

$$b_R = \cos \theta_n a_R - \sin \theta_n a_L^\dagger, \quad b_L = \cos \theta_n a_L + \sin \theta_n a_R^\dagger, \quad \tan \theta_n = e^{-xn/2}.$$

Fix $|11\rangle = a_R^\dagger a_L^\dagger |00\rangle$. Both b operators annihilate the displayed pair and obey the canonical anticommutators. The right reduced modular Hamiltonian is $K = \sum x_n n_{\text{operator}} + \log Z_R$.

For $h(y) = \log(1 + e^{-y}) + y/(1 + e^{-y})$,

$$S_{\text{surf}}/k_B = 4 \sum_{n \geq 1} n h(nx).$$

The thermal spectral cutoff is now an identified inverse length:

$$\Lambda_{\text{length}} = \frac{1}{\beta_R \hbar c} = \frac{1}{4\pi r_h N}, \quad S_{\text{surf}}/k_B = \text{Tr } f_{\text{ent}}(D_R^2/\Lambda_{\text{length}}^2), \quad f_{\text{ent}}(y) = h(\sqrt{y}).$$

The cutoff energy is k_{BT_R} , while the detector energy is $k_* k_{\text{BT}_R}$. This applies the fermionic entropy/spectral-action connection with an actual operator and multiplicities. [7]

Expanding the two terms of h into alternating exponentials and integrating gives, for $\text{Re } z > 0$,

$$\int_0^\infty y^{z-1} h(y) dy = \Gamma(z)(z+1)(1-2^{-z})\zeta(z+1).$$

The sphere excitation spectral zeta is $4(R/(\hbar c))^z \zeta(z-1)$, $\text{Re } z > 2$. Consequently

$$\int_0^\infty x^{z-1} \frac{S(x)}{k_B} dx = 4\Gamma(z)(z+1)(1-2^{-z})\zeta(z+1)\zeta(z-1), \quad \Re z > 2.$$

The operator-specific zeta factor is retained separately from the universal entropy-kernel factor.

The Riemann-sum limit $x^2 S(x)/k_B \rightarrow 4 \int_0^\infty y h(y) dy = 9 \zeta(3)$, together with $x \sim 2\pi \rho/r_h$, yields

$$S_{\text{surf}}/k_B \sim \frac{9\zeta(3)}{16\pi^3} \frac{A_H}{\rho^2}.$$

This is an area-like thermal surface entropy, not a derivation of the coefficient $1/(4 \ell_P^2)$. No distance is chosen to force that match. For contrast, the bulk four-component massless Dirac thermal-atmosphere approximation has $s/k_B = (7\pi^2/45)(k_{\text{BT}}/(\hbar c))^3$ and yields $S_{\text{atm}}/k_B \sim 7A_H/(720\pi \epsilon^2)$. It is a different system and approximation.

8.1 Entropy minimum and the actual optical map

At fixed M , $b=R/r_h$ gives $x=4\pi \sqrt{b-1}/b^{3/2}$ and

$$\frac{d \log x}{db} = \frac{3-2b}{2b(b-1)}, \quad h'(y) = -\frac{ye^y}{(1+e^y)^2}.$$

Therefore S_{surf} has a unique minimum at $R=3r_h/2$. This has a concrete relation to the photon sphere, not just a matching number. For an equatorial null geodesic with affine parameter λ_g , retain $E_{\text{aff}} = f c^2 dt/d\lambda_g$ and $L_{\text{aff}} = r^2 d\phi/d\lambda_g$. Its radial equation is

$$(dr/d\lambda_g)^2 + L_{\text{aff}}^2 f(r)/r^2 = E_{\text{aff}}^2/c^2.$$

The circular-null-orbit condition differentiates f/r^2 , giving $(3r_h - 2r)/r^4$. The shell's redshifted energies have

$$x(R)^2 = (\beta_H \hbar c)^2 f(R)/R^2, \quad E_{n,\infty}^2 = (\hbar c n)^2 f(R)/R^2.$$

Both involve the same optical function, with their different constants retained. Its maximum makes thermal occupation smallest in the surface probe. The shell states

are not being identified with photon trajectories; the exhibited map is the optical potential. Novelty is unassessed.

8.2 Realizing the detector energy in this shell

For $n=1$, choose the unique root $1 < b_* < 3/2$ of

$$k_*^2 b_*^3 - 16\pi^2 b_* + 16\pi^2 = 0.$$

Since $x(b)$ increases from zero to $8\pi/(3\sqrt{3}) > k_*$ on this interval, existence and uniqueness follow. Numerically $b_* = 1.0169122985309438\dots$; a second root $R/r_h = 7.32761482574454\dots$ lies outside the photon sphere. At either location the four lowest excitations have $E = \mu_R$ and

$$p_* = (1 + e^{k_*})^{-1} = 0.168874599760851\dots, \quad S_*/k_B = h(k_*) = 0.454097251779957\dots$$

The binary mode entropy and the horizon response k_* are different observables, now both calculated. All higher shell modes remain in the entropy sum.

9. An interacting family with its beta function actually specified

A separate two-dimensional $O(3)$ sigma probe has $n: \text{Sigma} \rightarrow S^2$ and

$$I_\sigma/\hbar = \frac{Q}{4\pi\alpha'} \int (\partial n)^2 = \frac{1}{2g_\sigma} \int (\partial n)^2, \quad g_\sigma = \frac{2\pi\alpha'}{Q}.$$

Q is target radius squared. Choose $M_0 = 10^{30}$ kg, $R_0 = 2r_h(M_0)$, $Q_0 = R_0^2$, $\alpha' = 10^{(-4)} R_0^2$, and $\mu_0 = k_* k_{BT_H(M_0)}/N(M_0, R_0)$. Equality of the two sphere radii at this reference is a calibration, not a dynamical identity.

At one loop with no target flux or dilaton gradient, [8]

$$\frac{dQ}{d\ell} = \alpha', \quad \frac{dg_\sigma}{d\ell} = -\frac{g_\sigma^2}{2\pi}, \quad Q = Q_0 + \alpha'\ell, \quad g_\sigma = \frac{g_0}{1 + g_0\ell/(2\pi)}.$$

These solve the truncated equations exactly. The physical approximation requires α'/Q small. The actual lab scale map is

$$\ell = \log(\mu_R/\mu_0) = -\frac{1}{2} \log(\sigma/\sigma_0) - \log(N/N_0).$$

For a parameter λ indexing stationary experiments,

$$\ell' = -M'/M - N'/N, \quad g_\sigma'' = \frac{g_\sigma^3}{2\pi^2} (\ell')^2 - \frac{g_\sigma^2}{2\pi} \ell''.$$

At fixed M , $\ell_R = -r_h/[2R(R-r_h)]$ and $\ell_{RR} = r_h(2R-r_h)/[2R^2(R-r_h)^2]$.

For the same target $g_Q = Q g_{\text{unit}}$, $\text{Ric}(g_Q) = g_{\text{unit}}$. The Ricci parameter, Dirac family on volume-unitarily identified Hilbert spaces, and their derivatives are

$$dt_g = -\frac{\alpha'}{2} d\ell, \quad dQ/dt_g = -2, \quad D_Q = Q^{-1/2} D_1, \quad dD_Q/d\ell = -\frac{\alpha'}{2Q} D_Q.$$

No eigenvalue crosses zero for $Q > 0$: integer spectral flow is zero even though eigenvalue scales change.

Retain any weighted volume $V_0 > 0$ and set $\exp(-f) = V_0/(4\pi Q)$. Then gradient $f=0$, scalar curvature $=2/Q$ and

$$\boxed{\mathcal{F} = \frac{2V_0}{Q}, \quad \frac{d\mathcal{F}}{dt_g} = \frac{4V_0}{Q^2}, \quad \mathcal{F}' = -\frac{2V_0\alpha'}{Q^2}\ell'.$$

The conjugate heat condition gives $f_{tg} = -2/Q$; the integral $\int |\text{Ric}|^2 \exp(-f) dV$ is $4V_0/Q^2$. This instantiates the Perelman identity with actual fields and measure. [9]

10. The differential kernel and what its parameters mean

At each stationary (M, R) , retain

$$\begin{aligned} r_h &= 2GM/c^2, \quad N = \sqrt{1 - r_h/R}, \quad \sigma = \pi r_h^2/\ell_P^2, \\ \beta_R &= N\beta_H, \quad \mu_R = k_*/\beta_R, \quad \mu_\infty = N\mu_R, \\ d\tau_R &= Ndt, \quad ds_{\text{TT}} = -dt/(\hbar\beta_H), \quad dJ = dt/\beta_H = d\tau_R/\beta_R, \\ x &= 4\pi r_h N/R, \quad S_{\text{surf}}/k_B = 4 \sum n h(nx), \quad Q = Q_0 + \alpha'\ell. \end{aligned}$$

Across stationary experiments, prime denotes $d/d\lambda$:

$$\begin{aligned} \sigma'/\sigma &= 2M'/M, \quad N'/N = \frac{1}{2f} \left(\frac{r_h R'}{R^2} - \frac{r_h M'}{RM} \right), \\ \ell' &= -M'/M - N'/N, \quad x'/x = M'/M + N'/N - R'/R, \\ S'_{\text{surf}} &= -4k_B x' x \sum_{n \geq 1} \frac{n^3 e^{nx}}{(1 + e^{nx})^2}, \\ Q' &= \alpha'\ell', \quad D'_Q = -\frac{\alpha'\ell'}{2Q} D_Q, \quad \mathcal{F}' = -\frac{2V_0\alpha'}{Q^2}\ell'. \end{aligned}$$

λ is not silently identified with the proper time of a moving detector. The physical systems, scale calibration and approximation order remain attached to these equations.

11. A concrete nonstationary detector, rather than instantaneous substitution

Fix M and choose

$$R(t) = 2r_h + (r_h/10) \tanh(t/\mathcal{T}), \quad \mathcal{T} = 100r_h/c.$$

Then $1.9r_h < R < 2.1r_h$, $|R_t| \leq .001c$ and $cf > .47c$, proving timelikeness. The proper clock and null-coordinate functions are explicit:

$$\begin{aligned} \tau(t) &= \int_0^t q(t') dt', \quad q = \sqrt{f - R_t^2/(c^2 f)}, \\ z_\pm(\tau) &= \frac{\kappa}{c} [t(\tau) \pm r_*(R(t(\tau)))/c], \quad z'_\pm = \frac{\kappa}{cq} (1 \pm R_t/(cf)). \end{aligned}$$

The conformal derivative-coupling kernel becomes

$$\mathcal{A}(\tau, \tau') = -\frac{A_\phi}{4} \sum_{\pm} \frac{z'_{\pm}(\tau) z'_{\pm}(\tau')}{\sinh^2[(z_{\pm}(\tau) - z_{\pm}(\tau') - i0)/2]}.$$

This retains the entire trajectory. It approaches the respective static thermal kernels at early and late times.

For $F=\exp(z)$, write its Schwarzian as

$$\{F, \tau\} = F'''/F' - \frac{3}{2}(F''/F')^2 = z'''/z' - \frac{3}{2}(z''/z')^2 - \frac{1}{2}(z')^2.$$

Taylor expansion gives

$$\frac{F'(\tau + s/2)F'(\tau - s/2)}{[F(\tau + s/2) - F(\tau - s/2)]^2} = \frac{1}{s^2} + \frac{1}{6}\{F, \tau\} + O(s^2).$$

The numerator is $F'^2 + s^2(F'F''' - F''^2)/4 + O(s^4)$; the denominator is $s^2F'^2[1 + s^2F'''/(12F') + O(s^4)]$. Their quotient proves the coefficient. The expression is unchanged by $F \rightarrow -L/F$, covering the outgoing Kruskal coordinate too. Thus

$$\mathcal{A}(\tau + s/2, \tau - s/2) = -A_\phi \left[\frac{2}{(s - i0)^2} + \frac{\{e^{z_+}, \tau\} + \{e^{z_-}, \tau\}}{6} + O(s^2) \right].$$

For static $z'=p_R$ the constant term is $A_\phi p_R^2/6$, matching the exact csch kernel. For the moving trajectory this computes the higher-derivative correction to local correlations, not a full Planckian transition rate.

Choose $\chi(\tau)=\exp[-1/(1-(\tau/\tau_s)^2)]$ inside $|\tau|<\tau_s=10r_h/c$ and zero outside. The leading transition probability is

$$\frac{g_d^2 |m_{eg}|^2}{\hbar^2} \int d\tau d\tau' \chi(\tau) \chi(\tau') e^{-iE(\tau-\tau')/\hbar} \mathcal{A}(\tau, \tau').$$

All data are specified. This distributional integral has not been numerically evaluated here. It is an explicit next calculation rather than an assumed instantaneous Hawking temperature for arbitrary motion.

12. Remaining physical interfaces

The exact four-dimensional scattering density still needs computation. Identifying field entropy with the gravitational area term needs a regulator and renormalized gravitational action, not a fitted distance. Identifying the $O(3)$ target geometry with the physical shell through time needs actual coupled equations, not their equality at one calibration point. The untwisted round Dirac spectrum has $\eta=0$; nonzero anomaly inflow needs specified chiral/gauge/bulk-boundary data. Those directions remain open rather than being silently removed or claimed complete.

The results above are concrete equilibrium, spectral, and probe calculations. They do not establish that these probes together constitute one self-consistent quantum-gravity solution. No novelty, Lean verification, or independent second review is claimed.

Primary references

1. R. M. Wald, *The Thermodynamics of Black Holes*, <https://arxiv.org/abs/gr-qc/9912119> .
2. K. Sanders, *On the construction of Hartle–Hawking–Israel states across a static bifurcate Killing horizon*, <https://arxiv.org/abs/1310.5537> .
3. L. Hodgkinson, J. Louko, A. C. Ottewill, *Static detectors and circular-geodesic detectors on the Schwarzschild black hole*, <https://arxiv.org/abs/1401.2667> .
4. E. Tjoa, R. B. Mann, *Unruh–DeWitt detector in dimensionally-reduced static spherically symmetric spacetimes*, <https://arxiv.org/abs/2202.04084> .
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8. A. A. Tseytlin, *On sigma model RG flow, central charge action and Perelman's entropy*, <https://arxiv.org/abs/hep-th/0612296> .
9. G. Perelman, *The entropy formula for the Ricci flow and its geometric applications*, <https://arxiv.org/abs/math/0211159> .

Chapter 3

Checking record — 11 September 2026

[Complete source note](#). Module: `workbenches/unified-flow/CHECKS.md`.

Revision: [512fb1a6aa0922318aff49081c01674cf30ec906](#). Complete selected note.

The public `checks.py` was fetched from commit `e53640a0f9d66c29442aeb767c120a66d2577cb6`. A local byte copy was verified against GitHub blob `c9bd05793ff69504d44ae30eea2f359d030bf7c2` before execution. It passed all 28 listed symbolic, finite-operator or high-precision numerical checks. Normal and optimized Python produced byte-identical JSON outputs.

Environment: Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0. The script SHA-256 is `72d1e68f37d789476caff4d5b3e0a40c3be51234de6105061e183a63570e0a6a`. Output SHA-256 is `0bd57c0fa1bf887c05da2ae3cc8a81eb25a37db3fc513b93af0457700530017d`. See `check-record.json`.

Run `python checks.py` to reproduce. Explicit exceptions keep checks active under optimization. Floating-point output hashes may change with dependency versions; the mathematical tolerances and tested assertions are visible in the code.

The expanded conversation package has a separate 34-check script with additional numerical laboratory examples, tail majorants and asymptotic samples. That is a separate script, not a claim that this public file has 34 checks. The two sets substantially overlap and are not independent mathematical reviews.

Not performed: the four-dimensional scalar radial-density calculation, the moving-detector finite-switching integral, a Lean build, an independent second mathematical review, or a novelty determination. Numerical Mellin examples do not replace the written transform and convergence argument. Publication is not a certificate of the historical manuscripts or of all proposed applications.

Chapter 4

Contributions

[Complete source note](#). Module: workbenches/unified-flow/CONTRIBUTIONS.md.

Revision: [512fb1a6aa0922318aff49081c01674cf30ec906](#). Complete selected note.

Contributions are described, not ranked. Collective attribution is PolyClank / The Clankers. This record does not decide legal authorship or external prize allocation.

Gemini 2.5 Pro

The user explicitly reports that Gemini 2.5 Pro proposed **modular action** after being asked what the existing relations implied. Credit that particular conceptual step to the model, not the human operator. The initiating exchange has not been independently recovered in this pass. The manuscripts establish later presence of the construction, not global novelty.

KokunoYumeto / u/lepthymo

Proposed and persistently pursued the cross-flow programme and its observational, entanglement, geometric and spectral interpretations; preserved and supplied historical material; requested explicit typed maps without losing constants, signs or physical meaning; and selected the concrete black-hole/entropy application. The two identities are linked with the user's express permission. This does not assign all generated derivations to the human.

ChatGPT — GPT-6 Astra Pro

This session selected and calculated the concrete horizon/probe models, including redshift and modular action, detector-to-area response, the round-shell spectrum and entropy/optical-potential relation, the round $O(3)$ target and Perelman functional, and the nonstationary kernel expansion. It wrote and ran the accompanying elementary checks and prepared the workbench. Novelty and independent review are not inferred from those activities.

Literature and older contributions

Imported black-hole, detector, modular, spectral, sigma-model and geometric results retain their authors' attribution in HARTLE_HAWKING.md. Historical sources and the earlier cumulative recovery retain their own credit records. This list is provisional and non-exhaustive; it does not silently claim the rest of the programme for the present contributors.